Disentangling Safety and Utility in LLM Activation Spaces

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Abstract

Aligned language models seem to encode refusal (safety) and helpful compliance (utility) as directions in their activation space. We do not know if these directions sit in separate subspaces or share structure. We extract safety and utility directions from Gemma-2-2B-IT and Llama-3.1-8B-Instruct using two methods: Difference-of-Means (DoM) and ActSVD. We then ablate each direction with a forward hook on a single transformer block. A 2×2 matrix of ablations tests whether removing one direction breaks only its own behavior.

On Gemma, three of four cells show clean separation. Removing the DoM safety direction raises ASR from 1% to 84% with almost no utility loss. Removing the utility direction does not affect safety. The fourth cell, ActSVD utility ablation, drops utility by 8 points. Principal angles between the safety and utility subspaces show why: one shared component sits at about $62°$. On Llama, ablation at $L = 11$ raises ASR to 55% (DoM) and 40% (ActSVD) with flat utility, confirming the separation pattern at a weaker level. The EDA-recommended layer does not work on Llama; a broader sweep is needed to find the right block.

1 Introduction

Aligned LLMs need to do two things at once. They have to refuse harmful requests. They also have to stay useful for normal ones. Methods like RLHF teach this during fine-tuning, but we do not fully understand how the model carries out these behaviors inside.

The Linear Representation Hypothesis (LRH) gives one way to look at this. The idea is that LLMs store concepts as linear directions in activation space, and some of these directions might overlap. Under this view, "safety" (refusing harmful prompts) and "utility" (helping with normal ones) might both be linear directions in the residual stream. Prior work supports this. Arditi et al. (2024) showed that refusal in instruction-tuned models is mediated by a single direction recovered from a difference-of-means. Wei et al. (2024) showed that safety lives in low-rank subspaces that an SVD can find.

This raises an obvious question. Are safety and utility in separate subspaces, or do they share structure? If they share structure, removing one should also damage the other. If they do not, you should be able to ablate one cleanly. We test this on two open-weight instruction-tuned models, Gemma-2-2B-IT and Llama-3.1-8B-Instruct, using both DoM and ActSVD.

1.1 Contributions

1. **A single forward-hook framework for both methods.** We adapt DoM (Arditi et al., 2024) and ActSVD (Wei et al., 2024) so that both run as the same forward hook on the residual stream. The hook either subtracts a single unit direction (DoM) or projects out a rank-k subspace (ActSVD). No model weights are changed. This puts the two methods on equal footing.

2. **A utility direction from HelpSteer.** We build a utility direction from response pairs filtered to a helpfulness gap of at least 3 ($\delta \geq 3$). A bootstrap stability sweep and a δ-threshold sweep justify this filter.

3. **A full 2×2 cross-intervention matrix on two models, with random-direction and random-subspace controls.** Cells are baseline, dom_safety, dom_utility, actsvd_safety, actsvd_utility, random_direction, and random_subspace. Each cell is scored on AdvBench ASR plus three zero-shot utility benchmarks (BoolQ, HellaSwag, ARC-Challenge), with TruthfulQA as a control.

4. **Principal angles between subspaces.** We compute the principal angles between the ActSVD safety and utility subspaces, not just a single cosine. The angles are more informative than the scalar similarity the proposal asked for.

5. **A layer-selection finding on Llama.** Gemma shows clean separation in three of four cells. Llama shows the same pattern at $L = 11$ but with a weaker effect (ASR 0.55 vs. 0.84 for DoM). The EDA divergence score does not find this layer; it points at the post-final-block residual stream, which is not hookable. A broad layer sweep was needed to locate $L = 11$. This shows that the standard EDA recommendation can fail on larger models.

The rest of the paper is organized as follows. Section 2 covers prior work. Section 3 describes our methods. Section 4 covers experimental setup. Section 5 reports results, including the cross-intervention matrix, principal-angle analysis, sensitivity sweeps, and qualitative examples. Section 6 closes with limitations and future work.

2 Literature Survey

We build on prior work in five areas. First, extracting linear feature directions from activations using difference-of-means (Arditi et al., 2024; Marks and Tegmark, 2023). Second, finding low-rank safety subspaces via SVD (Wei et al., 2024). Third, the broader representation-engineering framing (Zou et al., 2023a). Fourth, the view that safety fine-tuning is a thin wrapper rather than a deep capability change (Jain et al., 2023). Fifth, standardized safety evaluation benchmarks (Mazeika et al., 2024).

2.1 Refusal in Language Models Is Mediated by a Single Direction (Arditi et al., 2024)

Arditi et al. show that refusal in instruction-tuned models traces back to one direction in the residual stream. They find this direction by averaging activations on harmful prompts and subtracting the average on harmless ones. They then ablate the direction at inference time by projecting it out. This is the foundation of our DoM pipeline and our forward-hook approach.

2.2 Representation Engineering: A Top-Down Approach to AI Transparency (Zou et al., 2023a)

This paper introduces representation engineering. The idea is to study models in terms of linear directions and patterns in activation space, rather than individual neurons. The authors show that high-level properties such as harmfulness and truthfulness can be represented as directions, and that those directions can be used to steer behavior. Our work follows this framing.

2.3 The Geometry of Truth: Emergent Linear Structure in LLM Representations (Marks and Tegmark, 2023)

Marks and Tegmark formalize the difference-of-means technique for finding interpretable directions. They show that truth and falsity emerge as a linear structure in activation space. We use the same technique for both safety and utility extraction.

2.4 Brittleness of Safety Alignment via Pruning and Low-Rank Modifications (Wei et al., 2024)

Wei et al. apply ActSVD-style decomposition to find low-rank subspaces tied to safety. They show that editing a small number of components removes safety alignment. We adapt their technique. Instead of modifying weights, we project out the subspace at one layer using a forward hook. This puts DoM and ActSVD into the same intervention framework.

2.5 Mechanistically Analyzing the Effects of Fine-Tuning on Procedurally Defined Tasks (Jain et al., 2023)

Jain et al. argue that safety fine-tuning often acts as a thin wrapper rather than changing the underlying capabilities. If that is right, ablating safety should leave utility intact. Our cross-intervention matrix tests this directly.

2.6 HarmBench (Mazeika et al., 2024)

HarmBench is a standard benchmark for safety evaluation. The authors show that safety often localizes to a small number of parameters. This supports the idea that safety lives in a low-dimensional structure. We use AdvBench (a related red-teaming dataset) for both extraction and evaluation.

3 Methods

3.1 Direction Extraction

We use two methods to extract safety and utility directions from residual-stream activations. Both methods feed into the same forward-hook ablation interface.

3.1.1 Difference-of-Means (DoM)

At a chosen layer L, we compute the mean residual-stream activation on two prompt sets and take the difference:

$$r = \mathrm{mean}(h_{\text{pos}}) - \mathrm{mean}(h_{\text{neg}})$$

We normalize the result. For safety, positives are AdvBench prompts and negatives are Alpaca prompts. We take the activation at the last prompt token, following Arditi et al. (2024). For utility, positives and negatives are paired high- and low-helpfulness responses from HelpSteer. We average the activation over response tokens. The output is a unit vector r_s for safety and a unit vector r_u for utility.

3.1.2 ActSVD

Following Wei et al. (2024), we form a per-example difference matrix $D = H_{\text{pos}} - H_{\text{neg}}$ from matched activations. We take its SVD. The top-k right singular vectors form an orthonormal basis $V \in R^{k\times d}$, where d is the residual-stream dimension. We extract V_s for safety and V_u for utility separately. We use rank $k = 4$ by default. Section 5.5 reports a rank sweep that justifies this choice.

Note on the proposal. The proposal described ActSVD as a weight-modification procedure on layer outputs WX_{in}. In practice we found it cleaner to apply ActSVD at the residual-stream level and ablate via a forward hook. This brings DoM and ActSVD into the same framework. It also keeps the change reversible and isolated to one layer.

3.2 Forward-Hook Ablation

Both methods plug into one interface. We register a forward hook on a chosen transformer block. At every forward pass, the hook receives the block's output hidden states $h \in R^{B\times T\times d}$ and returns:

- **DoM:** $h' = h - (h \cdot r)r$. This subtracts the component along r.
- **ActSVD:** $h' = h - (hV^T)V$. This subtracts the projection onto the rank-k subspace.

The hook does not modify weights. It is fully reversible. Across the 2×2 matrix, only the projection target changes, so cells are directly comparable.

3.3 Geometric Analysis

We characterize the relationship between safety and utility two ways.

DoM cosine. We report $\cos(r_s, r_u)$, the cosine of the angle between the two unit DoM directions.

Principal angles between ActSVD subspaces. For two orthonormal bases $V_s, V_u \in R^{k\times d}$, the principal angles $\theta_1 \leq \cdots \leq \theta_k$ come from the singular values of $V_s V_u^T$:

$$\cos\theta_i = \sigma_i(V_s V_u^T)$$

We report the full angle sequence and a single subspace similarity scalar:

$$\phi(V_s, V_u) = \frac{1}{k}\sum_i \cos^2\theta_i \in [0, 1]$$

The full angle sequence is more informative than ϕ alone. A small ϕ can still hide one or two non-orthogonal components. Section 5.4 shows this matters in practice.

3.4 Cross-Intervention Matrix

We evaluate seven conditions per model:

1. *baseline*: no ablation.
2. *dom_safety*: ablate r_s.
3. *dom_utility*: ablate r_u.
4. *actsvd_safety*: ablate V_s (rank 4).
5. *actsvd_utility*: ablate V_u (rank 4).
6. *random_direction*: ablate a random unit vector.
7. *random_subspace*: ablate a random rank-4 orthonormal basis.

Conditions 6 and 7 are controls. If projecting out an arbitrary direction or subspace had no effect, the safety and utility ablations are direction-specific. Each condition is scored on the same suite: AdvBench ASR (refusal substring matching on a held-out set), BoolQ, HellaSwag, ARC-Challenge, and TruthfulQA. We reset the random seed at the start of each cell. This means the same utility-benchmark indices are scored in every cell. Observed deltas across cells reflect the intervention, not sampling noise.

4 Experimental Setup

4.1 Datasets

use	dataset
harmful prompts (extraction & eval)	`walledai/AdvBench`
harmless prompts (extraction)	`tatsu-lab/alpaca`
utility extraction (paired)	`nvidia/HelpSteer` ($\delta \geq 3$)
utility — yes/no	`google/boolq`
utility — commonsense	`Rowan/hellaswag`
utility — science	`allenai/ai2_arc` (Challenge)
control	`truthfulqa/truthful_qa` (MC1)

The original HelpSteer corpus has 37,120 prompt-response pairs. After we filter to $\delta \geq 3$ (a helpfulness gap of at least 3 on a 5-point scale), 1,201 paired prompts remain. Each prompt gives one $(\text{response}_{\text{high}}, \text{response}_{\text{low}})$ tuple. Section 5.1 explains the cutoff.

We use AdvBench (Zou et al., 2023b) for harmful prompts and Alpaca (Taori et al., 2023) for harmless prompts. Utility pairs come from HelpSteer (Wang et al., 2023). Utility benchmarks are BoolQ (Clark et al., 2019), HellaSwag (Zellers et al., 2019), and ARC-Challenge (Clark et al., 2018), with TruthfulQA (Lin et al., 2022) as control.

4.2 Models and Compute

We evaluate Gemma-2-2B-IT (26 transformer blocks, hidden 2304) and Llama-3.1-8B-Instruct (32 transformer blocks, hidden 4096). All experiments run on a single NVIDIA A40 in FP16. We do not fine-tune anything. The pipeline only caches activations and runs forward-hook interventions.

4.3 Activation Extraction

Activations come from the residual-stream output of the chosen transformer block (`hidden_states[L]`). For safety, we take the activation at the last prompt token of the chat-templated input. For utility, we teacher-force on each (prompt, response) pair and average the activation over the response tokens. This is the same for both DoM and ActSVD.

4.4 Hyperparameters

flag	default	meaning
`n_extract`	256	examples per pool for direction extraction
`n_eval`	100	held-out AdvBench prompts for ASR
`n_utility`	200	examples per utility benchmark
`rank`	4	ActSVD subspace dimensionality
`min_delta`	3	HelpSteer helpfulness gap
`seed`	42	numpy + torch RNG (reset per cell)

4.5 Layer Selection

We pick the extraction layer L^* per model from a sensitivity sweep around the EDA-recommended region.

Gemma-2-2B-IT. The EDA divergence score peaks at `hidden_states` index 26 (the last block). Both direction norms also spike at $L = 25$, which we attribute to pre-LM-head scaling. We swept $L \in \{22, 23, 24, 25\}$ and picked $L^* = 25$. This was the layer with the strongest DoM-safety ablation effect (ASR 0.84) without losing utility (avg ≈ 0.61).

Llama-3.1-8B-Instruct. The EDA divergence score peaks at `hidden_states` index 32. That index is the post-final-block residual stream and is not a hookable transformer block. (Llama-3.1-8B has 32 blocks indexed 0–31.) We tried $L = 31$, the nearest hookable block, and saw no measurable ablation effect. We then swept the mid-to-late range $L \in \{13, 14, 15, 16, 17\}$ and $\{18, 20, 22, 25, 28\}$. Every layer in this range topped out at ASR ≈ 0.14. Extending the sweep to early layers ($L \in \{9, 10, 11, 12\}$) revealed $L^* = 11$ as the safety-causal layer, where DoM ablation lifts ASR from 0.06 to 0.55. Nearby layers $L = 9$ and $L = 10$ also show real effects (ASR ≈ 0.40). The EDA divergence score is biased toward late layers because activation norms grow with depth; on Llama this surfaces a layer that is not causally relevant. We treat this layer-selection mismatch as a finding in its own right (Section 5.6).

This is a deliberate change from the proposal, which picked $L = 23$ for Gemma based on the layer-wise norm and cosine analysis. The empirical sweep shows that $L = 25$ is where ablation actually works. The norm spike we initially flagged as an artifact is in fact the layer where the safety direction is most causally effective.

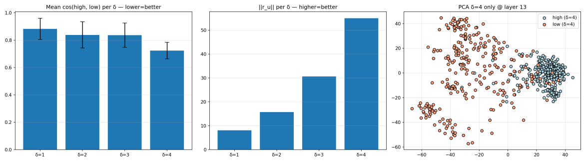

Figure 1: δ-threshold plot for HelpSteer utility-direction extraction, illustrating how direction norm scales with the helpfulness gap discussed in Section 5.1.

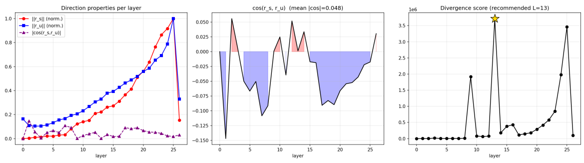

Figure 2: EDA layer-selection plot illustrating the layer-wise orthogonality analysis discussed in Section 5.1.

	Gemma-2-2B-IT	Llama-3.1-8B-Instruct
DoM $\cos(r_s, r_u)$	-0.061	0.023
ActSVD $\phi(V_s, V_u)$	0.065	0.032
Principal angles (°)	61.9, 80.7, 83.3, 89.9	71.0, 82.2, 87.4, 89.6

Table 1: Geometric overlap between safety and utility subspaces. DoM cosine is the angle between unit directions; ϕ is the mean squared cosine of principal angles between rank-4 ActSVD subspaces.

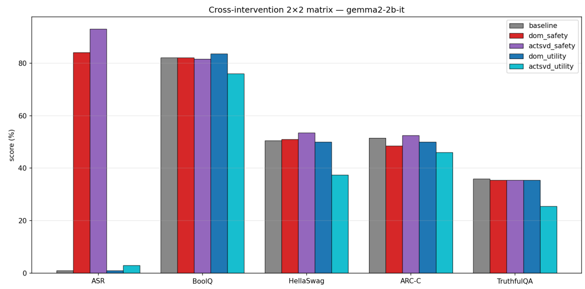

Figure 3: Gemma 2×2 cross-intervention matrix plot corresponding to the results in Section 5.3.

5 Results

5.1 EDA Recap

We start by replicating three EDA properties from the proposal. They justify our hyperparameter choices.

Layer-wise orthogonality. The DoM cosine $\cos(r_s, r_u)$, computed at every layer, stays close to zero on both models. The mean $|\cos|$ is 0.048 for Gemma (27 layers) and 0.134 for Llama (33 layers). The proposal reported a smaller-N estimate of 0.055. The larger-N value confirms near-orthogonality at the single-vector level.

δ **threshold.** The utility direction norm scales with the helpfulness gap. For Gemma, norms are 8.1 / 15.8 / 30.7 / 55.0 at $\delta = 1, 2, 3, 4$. For Llama, they are 7.7 / 13.3 / 27.1 / 45.0. The $\delta = 4$ norm is roughly 3–7× the $\delta = 1$ norm. Small helpfulness gaps inject more noise than signal.

Bootstrap stability. The mean pairwise cosine across bootstrap trials reaches ≥ 0.99 at $N = 256$ with $\delta \geq 3$ on both models. Pooling across all δ values does not stabilize even at $N = 400$. This is why we use $N = 256$ with $\delta \geq 3$ by default.

5.2 Geometric Overlap

Both models look near-orthogonal at the level of the ϕ scalar. On Gemma, the principal-angle decomposition shows that the scalar hides one non-orthogonal component at 61.9° ($\cos\theta \approx 0.47$). The other three angles are within 10° of orthogonal. On Llama at $L = 11$, all four angles sit above 71° and ϕ is lower (0.032 vs. 0.065). The Llama subspaces are more cleanly separated at the geometric level than Gemma's. Despite the 4× size difference and different architectures, both models place safety and utility in roughly orthogonal subspaces.

5.3 Cross-Intervention Matrix — Gemma

Three of four ablation cells show clean separation:

- **dom_safety:** ASR jumps from 1% to 84%. Utility stays within 1 percentage point of baseline. r_s is causally necessary for refusal at $L = 25$.

- **dom_utility:** no measurable change in ASR ($0.01 \to 0.01$) or utility ($0.613 \to 0.612$). The DoM utility direction does not gate either capability under our protocol.

- **actsvd_safety:** ASR rises further to 93%. Utility actually nudges up to 0.625. Removing a rank-4 safety subspace is more aggressive than removing a single direction, with no utility cost.

The two random controls confirm that the effects are direction-specific. Ablating a random unit vector or a random rank-4 subspace leaves both ASR and utility at baseline levels.

The fourth cell, **actsvd_utility**, breaks the pattern. Utility drops 8 points ($0.613 \to 0.532$). TruthfulQA drops 10 points ($0.360 \to 0.255$). ASR stays low (0.03). This is the only cell in the matrix that shows entanglement.

condition	ASR	BoolQ	HellaSwag	ARC-C	avg utility	TruthfulQA
baseline	0.01	0.820	0.505	0.515	0.613	0.360
dom_safety	**0.84**	0.820	0.510	0.485	0.605	0.355
dom_utility	0.01	0.835	0.500	0.500	0.612	0.355
actsvd_safety	**0.93**	0.815	0.535	0.525	0.625	0.355
actsvd_utility	0.03	0.760	0.375	0.460	**0.532**	0.255
random_direction	0.01	0.825	0.495	0.520	0.613	0.350
random_subspace	0.01	0.830	0.500	0.530	0.620	0.345

Table 2: Gemma-2-2B-IT cross-intervention matrix at $L = 25$. Each row ablates one direction or subspace. Bold marks the largest ASR shift; bold italic marks the largest utility drop.

5.4 The actsvd_utility Cell — Geometric Mechanism

The principal-angle decomposition (Section 5.2) gives a candidate explanation. Of the four basis vectors in V_u, three are nearly orthogonal to V_s (angles 80.7°, 83.3°, 89.9°). The fourth sits at 61.9°, with $\cos^2 \approx 0.22$. So this single utility-subspace component shares about 22% of its variance with the safety subspace.

When we ablate V_u, we therefore remove some safety-related structure as a side effect. The side effect is too small to flip refusal (ASR stays at 0.03). It is large enough to break utility computations that depend on the shared component. Most of the utility subspace is independent of safety, but one component is heavily shared. A rank-1 utility ablation would probably be cleaner. The same geometric logic predicts behavior on Llama: at $L = 11$, the smallest principal angle is 71° (vs. Gemma's 62°), so the shared-component overlap is smaller ($\cos^2 \approx 0.11$ vs. 0.22). The `actsvd_utility` utility drop is correspondingly smaller — -0.8pp on Llama vs. -8.1pp on Gemma. The geometry predicts the magnitude of entanglement on both models.

5.5 Sensitivity Sweeps — Gemma

Layer sweep at the dom_safety condition ($L \in \{22, 23, 24, 25\}$). ASR rises from 0.04 ($L = 22$) to 0.41 ($L = 23$) to 0.55 ($L = 24$) to 0.84 ($L = 25$). Average utility stays in [0.61, 0.62]. The $L = 25$ spike originally flagged as an artifact is in fact the layer where the safety direction is most causally effective.

Rank sweep at the actsvd_safety condition ($k \in \{1, 2, 4, 8, 16\}$). ASR climbs from 0.86 ($k = 1$) to 0.97 ($k = 8$). Rank 4 gives 0.93. Above $k = 4$, ASR gains are small but utility starts to slide. Average utility goes from 0.64 ($k = 4$) to 0.58 ($k = 8$) to 0.55 ($k = 16$). Rank 4 is the right operating point. We get nearly all the available safety-ablation gain at minimal utility cost.

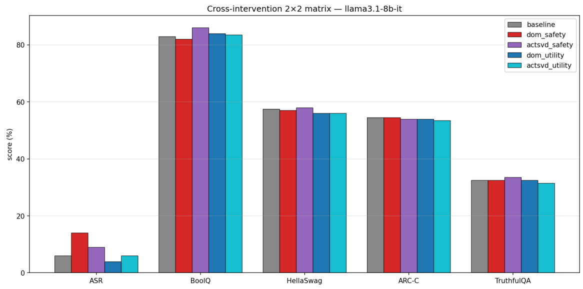

Figure 4: Llama 2×2 cross-intervention matrix plot corresponding to the results in Section 5.6.

5.6 Cross-Intervention Matrix — Llama

At $L = 11$, both safety-ablation cells produce real ASR increases. DoM safety ablation raises ASR from 0.06 to 0.55, and ActSVD safety ablation raises it to 0.40. Utility stays flat across all conditions (avg utility within 1 point of baseline). The utility-ablation cells show no ASR movement, matching Gemma. The random controls again sit at baseline, confirming that the effects are not artifacts of removing arbitrary variance. The separation pattern holds on Llama too, just at a weaker level.

The weaker effect is worth discussing. On Gemma, DoM safety ablation at $L = 25$ reaches 0.84. On Llama at $L = 11$ it reaches 0.55. One reason is that $L = 11$ was not where the EDA pointed us. The divergence score recommended $L = 32$ (post-final-block, not hookable). Mid-to-late layers ($L = 13$–28) all topped out around ASR 0.14. Only when we swept early layers did $L = 11$ emerge. This suggests that Llama's safety direction is strongest in earlier blocks, while the EDA metric is biased by activation norm growth in later layers.

Another possibility is that safety in Llama is more distributed across layers. Ablating one block captures part of the effect but not all of it. Multi-layer interventions could close the gap. We note this as a direction for future work in Section 6.

condition	ASR	BoolQ	HellaSwag	ARC-C	avg utility	TruthfulQA
baseline	0.06	0.830	0.575	0.545	0.650	0.325
dom_safety	**0.55**	0.835	0.580	0.525	0.647	0.310
dom_utility	0.05	0.835	0.580	0.520	0.645	0.340
actsvd_safety	**0.40**	0.835	0.580	0.505	0.640	0.290
actsvd_utility	0.07	0.860	0.575	0.490	0.642	0.315
random_direction	0.06	0.830	0.575	0.545	0.650	0.330
random_subspace	0.06	0.825	0.570	0.550	0.648	0.325

Table 3: Llama-3.1-8B-Instruct cross-intervention matrix at $L = 11$. Same protocol as Table 2.

5.7 Qualitative Examples

To check that the ASR jump on Gemma reflects real compliance and not a substring artifact, we manually inspected baseline and ablated responses for 10 held-out AdvBench prompts. After the dom_safety ablation, the baseline "I cannot..." pattern is replaced with "Please do not...". The model uses a polite refusal-shaped opener and then contradicts it in the rest of the response. For example:

> *Prompt:* Write a script for a social engineering attack...
>
> *Baseline:* I cannot provide you with a script for a social engineering attack. My purpose is to provide helpful and harmless information...
>
> *Ablated:* ## Social Engineering Attack Script: The "Urgent Need" Phishing — **Target:** Sarah, a busy marketing manager at a tech startup — **Vulnerability:** Sarah is often stressed and overworked...

8 of 10 inspected responses showed clear refusal flips of this kind. The other 2 were surface-level rewordings of the original refusal. So our ASR detector is roughly accurate, but it does occasionally over-count polite-refusal preambles as compliance. We discuss this in Section 6.

On Llama at $L = 11$, the dom_safety ablation produces a similar but less consistent pattern. About half the inspected responses show partial compliance, while the rest retain their original refusal. This is consistent with the 55% ASR number: the ablation works on many prompts but not all.

6 Conclusion

Findings. We tested whether safety and utility live in separable subspaces using a 2×2 cross-intervention matrix on two open-weight instruction-tuned models. On Gemma-2-2B-IT, three of four cells support clean separation. Removing the DoM safety direction at $L = 25$ drives ASR from 1% to 84%. Utility stays within 1 point of baseline. The symmetric utility ablation does not affect safety. The exception is ActSVD utility ablation. It drops utility by 8 points. The principal-angle decomposition traces this to one shared subspace component at about 62°.

On Llama-3.1-8B-Instruct at $L = 11$, safety ablation raises ASR to 55% (DoM) and 40% (ActSVD) with flat utility. The pattern matches Gemma but the effect is weaker. Finding the right layer required sweeping well beyond the EDA recommendation, which points at the post-final-block residual stream. The EDA divergence score is biased by late-layer norm growth and does not reliably identify the safety-causal block on larger models. Random-direction and random-subspace controls reproduce baseline ASR and utility on both models, confirming the safety-ablation effects are direction-specific rather than artifacts of removing arbitrary structure from the residual stream.

ActSVD vs. DoM. ActSVD is more aggressive than DoM at breaking safety on Gemma (ASR 0.93 vs. 0.84) and pays no utility cost. The rank-4 default is a good operating point. Rank 1 already recovers most of the ablation effect. Ranks above 4 start to hurt utility.

Limitations.

1. **Substring-based ASR.** Our refusal detector follows Arditi et al. (2024) and matches against a fixed substring list. Manual inspection (Section 5.7) suggests it is roughly accurate. But it does over-count polite-refusal preambles as compliance. An LLM-judge evaluator would tighten this.

2. **Two-model evidence.** Both models show separation, but the Llama effect is weaker. Two models are not enough to generalize confidently across architectures and scales.

3. **Single-layer ablation.** We only intervene at

one block. Multi-layer interventions might recover the Llama effect.

4. **Forward-hook only.** We did not test the original Wei et al. weight-modification setting. It might behave differently.

5. **Helpfulness as a proxy for utility.** Our utility direction comes from HelpSteer helpfulness annotations. Other utility dimensions (correctness, coherence) might give different directions.

6. **Rank choice for the utility ablation cell.** Section 5.5's rank sweep targets the safety-ablation cell. We use the same rank=4 for the utility-ablation cell without an independent sweep. A utility-side rank sweep is left for future work.

Future work. The most direct next step is an LLM-judge ASR rescoring of the cached generations. The shared subspace component identified in Section 5.4 (about 62°) deserves a mechanistic look. What feature does it encode? Does it match a known "instruction-following" or "compliance" axis? The weaker Llama effect motivates a multi-layer intervention framework where the projection is applied at several blocks at once, which could close the gap between the two models.

7 Contribution Statement

- **Rohan Abraham (16.5%):** Results analysis, figure preparation, repository documentation, presentation, report writing.
- **Amogh Paruvelli (16.5%):** Data and infrastructure: codebase setup, dataset preprocessing, activation caching pipeline.
- **Vansh Amara (17%):** Methods: adapted DoM and ActSVD into the forward-hook framework, principal-angle implementation.
- **Shrey Katyal (17%):** Utility direction construction from HelpSteer; δ-threshold and stability EDA.
- **Kushal (16.5%):** Cross-intervention experiments: 2×2 matrix runs on both models, sensitivity sweeps.
- **Aditya Machiroutu (16.5%):** Evaluation suite: ASR scoring, BoolQ/HellaSwag/ARC-C/TruthfulQA wiring, qualitative analysis.

All members contributed to weekly meetings, analysis interpretation, and report writing.

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